

SQL Queries

- Atliq Hardware Finance and Supply Chain Analytics Project

sold quantity, gross price per item, gross price total

```
SELECT
    s.date, s.product_code,

    p.product, p.variant, s.sold_quantity,

    g.gross_price as Gross_price_per_item,

    g.gross_price*s.sold_quantity as gross_price_total
```

```
FROM fact_sales_monthly s
```

```
JOIN dim_product p
```

```
ON
    p.product_code = s.product_code
```

```
JOIN fact_gross_price g
```

```
ON
    g.product_code = s.product_code  AND

    g.fiscal_year = get_fiscal_year(s.date)
```

```
WHERE
    customer_code=90002002  AND

    get_fiscal_year(date)=2021
```

```
ORDER BY date asc
```

```
Limit 1000000;
```

2Q. Total gross price per date

SELECT

monthname(s.date) as month,

SUM(ROUND(g.gross_price*s.sold_quantity,2)) as gross_price_total

FROM fact_sales_monthly s

JOIN fact_gross_price g

ON

g.product_code = s.product_code and

g.fiscal_year = get_fiscal_year(s.date)

WHERE

customer_code = 90002002

GROUP BY s.date

ORDER BY s.date asc;

3Q. Total gross price per FY

```
select
    g.fiscal_year,
    round(sum(s.sold_quantity * g.gross_price)/1000000,2)as
"gross_price_total(in mln)"

from fact_sales_monthly s

join fact_gross_price g

on
    s.product_code = g.product_code and
    get_fiscal_year(s.date) = g.fiscal_year

Where
    customer_code = 90002002

group by g.fiscal_year;
```

4Q. Top 5 product by net sales

```
SELECT
```

```
    Product,
```

```
    ROUND(SUM(net_sales)/1000000,2) AS net_sales_mln
```

```
FROM net_sales n
```

```
JOIN dim_customer c
```

```
ON
```

```
    n.customer_code=c.customer_code
```

```
WHERE fiscal_year=2021
```

```
    GROUP BY Product
```

```
    ORDER BY net_sales_mln desc
```

```
LIMIT 5;
```

#5Q. Top 5 Market by net sales

```
select
    market,
    round(sum(net_sales)/1000000,2) as net_sales_mln
from net_sales

Where
    fiscal_year = 2021

group by market

order by net_sales_mln desc

limit 5;
```

#6Q. Top 5 Customers by net sales

```
select
    c.customer,
    round(sum(net_sales)/1000000,2) as net_sales_mln
from net_sales s
join dim_customer c
using(customer_code)

Where
    fiscal_year = 2021

group by customer

order by net_sales_mln desc

limit 5;
```

7Q. Top 10 Customers by net_sales_% contribution

```
with cte1 as(

SELECT
    c.customer,

    ROUND(SUM(net_sales)/1000000,2) AS net_sales_mln

FROM net_sales n

JOIN dim_customer c

ON
    n.customer_code=c.customer_code

WHERE
    fiscal_year=2021

GROUP BY c.customer)

SELECT *,

    net_sales_mln*100/SUM(net_sales_mln) OVER() as pct

FROM cte1

ORDER BY net_sales_mln desc

Limit 10;
```


• # 8Q. Region wise net sales breakdown

with cte1 as(

SELECT

c.customer,c.region,

ROUND(SUM(net_sales)/1000000,2) AS net_sales_mln

FROM net_sales n

JOIN dim_customer c

ON

n.customer_code=c.customer_code

WHERE

fiscal_year=2021 AND region = "APAC"

GROUP BY c.customer,c.region)

SELECT *,

net_sales_mln*100/SUM(net_sales_mln) OVER() as pct_share_region

FROM cte1

ORDER BY region,net_sales_mln desc

9Q. Retrieve the top 3 products in every region by their total quantity in FY =2021

```
with cte1 as (select
                p.division,
                p.product,
                sum(sold_quantity) as total_qty

from fact_sales_monthly s

join dim_product p

On
    p.product_code=s.product_code
where
    fiscal_year =2021

group by p.product),

cte2 as(select *,

dense_rank() over(partition by division order by total_qty desc)as drnk
from cte1)

select * from cte2

Where
    drnk<=3
```

10Q. Supply Chain – forecast accuracy

```
with forecast_err_table as( Select

    s.customer_code,

    sum(s.sold_quantity) as total_sold_qty,

    sum(forecast_quantity-sold_quantity) as net_err,

    sum((forecast_quantity-sold_quantity))*100/sum(forecast_quantity) as net_err_pct,

    sum(abs(forecast_quantity-sold_quantity)) as abs_err,

    sum(abs(forecast_quantity-sold_quantity))*100/sum(forecast_quantity) as abs_err_pct

from gdb0041.fact_act_est s

where s.fiscal_year =2021

group by customer_code)

select  e.*,

        c.customer,c.market,

        if(abs_err_pct > 100, 0, 100-abs_err_pct) as forecast_accuracy

from forecast_err_table e

join dim_customer c

using(customer_code)

order by forecast_accuracy desc;
```